

Unit 1.1 : Numbers, intervals and inequalities

* Sets of numbers:

$\mathbb{N}$: set of all natural numbers

$\mathbb{Z}$: — || — integers

$\mathbb{Q}$: — || — rational numbers

$\mathbb{I}$: — || — irrational numbers

$\mathbb{R}$: — || — real numbers

* Interval notation: (see table 1, p A4)

* Inequality rules:

- (1) If $a < b$ then $a + c < b + c$
- (2) If $a < b$ and $c < d$ then $a + c < b + d$
- (3) If $a < b$ and $c > 0$ then $ac < bc$
- (4) If $a < b$ and $c < 0$ then $ac > bc$
- (5) If $0 < a < b$ then $\frac{1}{a} > \frac{1}{b}$

Example 1: Solve for x if $\frac{x+1}{2-x} \geq 0$

$\therefore$

$$\frac{x+1}{2-x} \geq 0 \Rightarrow \begin{matrix} x+1 \geq 0 \text{ and } 2-x > 0 & (1) \\ \text{or} \\ x+1 \leq 0 \text{ and } 2-x < 0 & (2) \end{matrix}$$

$$\Rightarrow x \geq -1 \text{ and } x < 2 \text{ OR } x \leq -1 \text{ and } x > 2$$

$$\Rightarrow -1 \leq x < 2 \text{ OR } \emptyset \text{ (empty set)}$$



$$\Rightarrow -1 \leq x < 2$$

or

$$x \in [-1, 2) \quad ([-1, 2) \cup \emptyset)$$

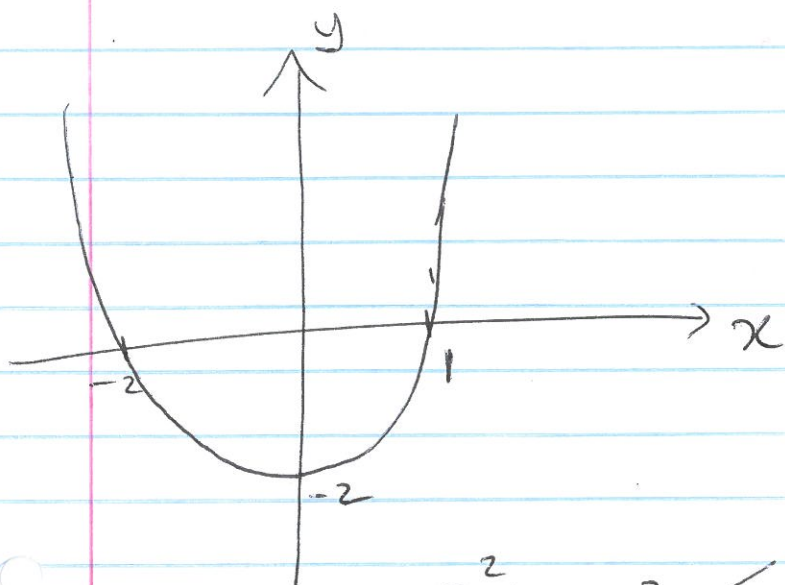
Example 2: Quadratic inequalities must be solve using a graph unless mentioned otherwise.

(a) Solve $x^2 + x < 2$

$$x^2 + x < 2 \Leftrightarrow x^2 + x - 2 < 0$$

$$(\text{observe with } g(x) = x^2 + x, \quad g(1) = 2 \text{ \& } g(-2) = 2)$$

$$x^2 + x - 2 = 0 \quad (\Rightarrow) \quad x = \frac{-1 \pm \sqrt{1 - 4(-2)}}{2}$$



$$x = \frac{-1 \pm 3}{2}$$

$$x \in \{-2, 1\}$$

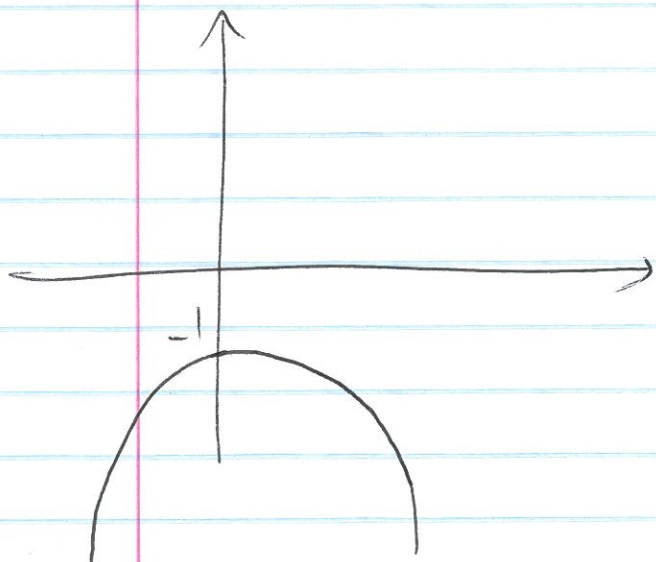
$$\text{or } x_1 = -2, x_2 = 1$$

$$x^2 + x - 2 < 0 \Rightarrow x \in (-2, 1)$$

NB: Always have details on your graph, especially intercepts with the axis.

$$(b) -x^2 - x - 1 < 0$$

Note that $x^2 + x + 1 = 0$ has no solutions (since $\Delta = 1 - 4(1) = -3 < 0$).



Hence $x \in \mathbb{R}$.

since $f(x) = -x^2 - x - 1 < 0$
for all $x \in \mathbb{R}$.

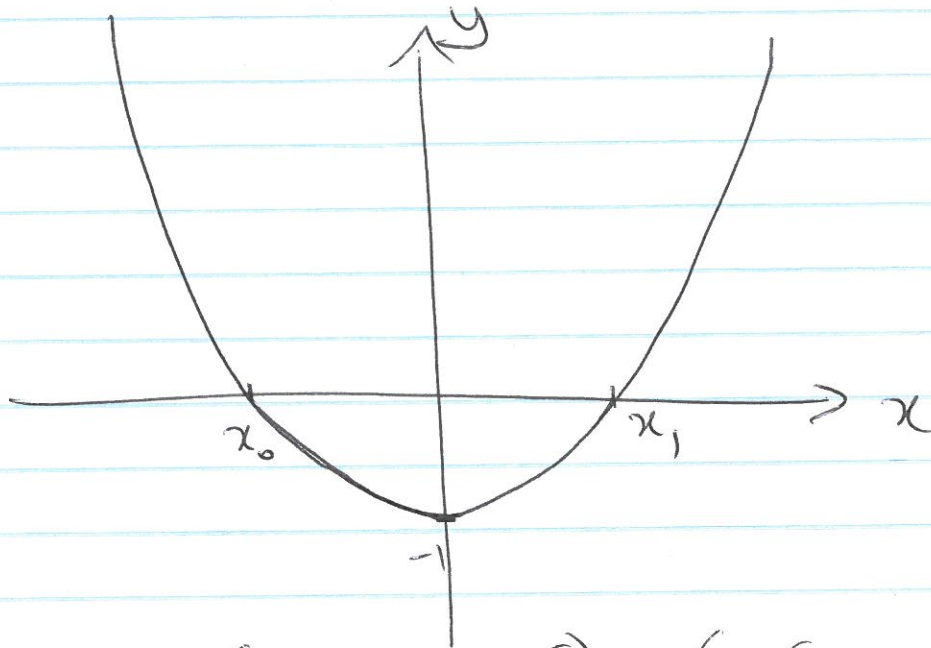
(c) solve algebraically

$$x^2 + x - 1 > 0$$

$$x^2 + x - 1 = 0 \quad \Leftrightarrow \quad x = \frac{-1 \pm \sqrt{5}}{2}$$

$$\Delta = 1 - 4 \times (-1) = 5$$

$$x_0 = \frac{-1 - \sqrt{5}}{2} (< 0) \text{ and } x_1 = \frac{-1 + \sqrt{5}}{2} (> 0)$$



$$x \in \left(-\infty, \frac{-1 - \sqrt{5}}{2}\right) \cup \left(\frac{-1 + \sqrt{5}}{2}, +\infty\right)$$